



SC ASSIGNMENT 4



TEAM MEMBER

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Data Dictionary for Course Registration System

Entity	Attribute	Data Type	Length	Description	Cardinality
CS Office	OfficeID	VARCHAR	10-15	Unique ID for the CS Office	1
Semester	SemesterID	INT	4-6	Unique ID for each Semester	1:M
	SemesterName	VARCHAR	10	Name of the Semester (Fall/Spring)	
Course	CourseID	VARCHAR	6-10	Unique Course Code	1:M
	CourseName	VARCHAR	50	Name Of Course	
	SemesterID	INT	4-6	Associated Semester	
Prerequisite	PrerequisiteID	VARCHAR	6-10	Unique ID of the prerequisite course	M:N
	CourseID	VARCHAR	6-10	The course requiring this prerequisite	
Course Schedule	ScheduleID	INT	4-6	Unique Schedule ID	1:M or 1:1
	CourseID	VARCHAR	6-10	Associated Course	
	Time		-	Time slot of the course	
	Day	VARCHAR	10	Day of the week	
Scheme of study	SchemeID	INT	4-6		1:M
	CourseID	VARCHAR	6-10	Associated Course	
	CourseName	VARCHAR	50	Name of the Course	
Student	StudentID	VARCHAR	10-15	Unique Student ID	M:N
	Name	VARCHAR	50	Full Name of the student	
	CourseID	VARCHAR	6-10	Registered Course	
Coordinator	CordinatorID	VARCHAR	10-15	Unique Coordinator ID	1:M
	Name	VARCHAR	50	Name of the Coordinator	
	Designation	VARCHAR	50	Position of the Coordinator	
	Type	VARCHAR	20	Role Type(Faculty/Admin)	
Registration Record	RegistrationID	VARCHAR	10-15	Unique Registration ID	1:M
	StudentID	VARCHAR	10-15	Registered Student ID	
	CourseID	VARCHAR	6-10	Registered Course Code	

Description

The **Data Dictionary** provides a structured definition of all entities and attributes within the **Course Registration System**. It serves as a reference for developers, database administrators, and stakeholders to understand how data is stored, processed, and related within the system.

This document includes details such as **attribute names, data types, length constraints, descriptions, and cardinality** (relationships between entities).

Entities and Their Attributes

1. CS Office

The **CS Office** manages the entire registration process, including handling semesters and courses.

- **OfficeID** (*VARCHAR, 10-15*) – Unique identifier for the CS Office.

Cardinality: One-to-Many (1:M) → A single CS Office manages multiple semesters.

2. Semester

Represents academic sessions such as **Fall** and **Spring** in which students enroll.

- **SemesterID** (*INT, 4-6*) – Unique identifier for a semester.
- **SemesterName** (*VARCHAR, 10*) – Name of the semester (e.g., Fall/Spring).

Cardinality: One-to-Many (1:M) → A single semester can have multiple courses.

3. Course

Represents the subjects offered in each semester.

- **CourseID** (*VARCHAR, 6-10*) – Unique course code.
- **CourseName** (*VARCHAR, 50*) – Full name of the course.
- **SemesterID** (*INT, 4-6*) – Semester in which the course is offered.

Cardinality: One-to-Many (1:M) → A semester can have multiple courses.

4. Prerequisite

Courses that a student must complete before enrolling in a higher-level course.

- **PrerequisiteID** (*VARCHAR, 6-10*) – Unique ID of the prerequisite course.
- **CourseID** (*VARCHAR, 6-10*) – The course that requires this prerequisite.

Cardinality: Many-to-Many (M:N) → A course can have multiple prerequisites, and a prerequisite can be required for multiple courses.

5. Course Schedule

Defines the timing and days for each course.

- **ScheduleID** (*INT, 4-6*) – Unique schedule identifier.
- **CourseID** (*VARCHAR, 6-10*) – The course associated with this schedule.
- **Time** (*TIME*) – The time slot for the course.
- **Day** (*VARCHAR, 10*) – The day of the week (e.g., Monday, Wednesday).

Cardinality: One-to-One (1:1) or One-to-Many (1:M) → A course may have one or more schedules.

6. Scheme of Study

A structured program defining which courses a student must take.

- **SchemeID** (*INT, 4-6*) – Unique identifier for the scheme of study.
- **CourseID** (*VARCHAR, 6-10*) – Course included in the scheme.
- **CourseName** (*VARCHAR, 50*) – Name of the course.

Cardinality: One-to-Many (1:M) → A scheme of study defines multiple courses.

7. Student

Represents students who register for courses.

- **StudentID** (*VARCHAR, 10-15*) – Unique identifier for each student.
- **Name** (*VARCHAR, 50*) – Full name of the student.
- **CourseID** (*VARCHAR, 6-10*) – Courses in which the student is enrolled.

Cardinality: Many-to-Many (M:M) → A student can enroll in multiple courses, and each course can have multiple students.

8. Coordinator

Faculty or administrative staff responsible for managing course enrollments and student records.

- **CoordinatorID** (*VARCHAR, 10-15*) – Unique identifier for a coordinator.
- **Name** (*VARCHAR, 50*) – Full name of the coordinator.
- **Designation** (*VARCHAR, 50*) – Position or title of the coordinator.
- **Type** (*VARCHAR, 20*) – Role type (e.g., Faculty, Admin).

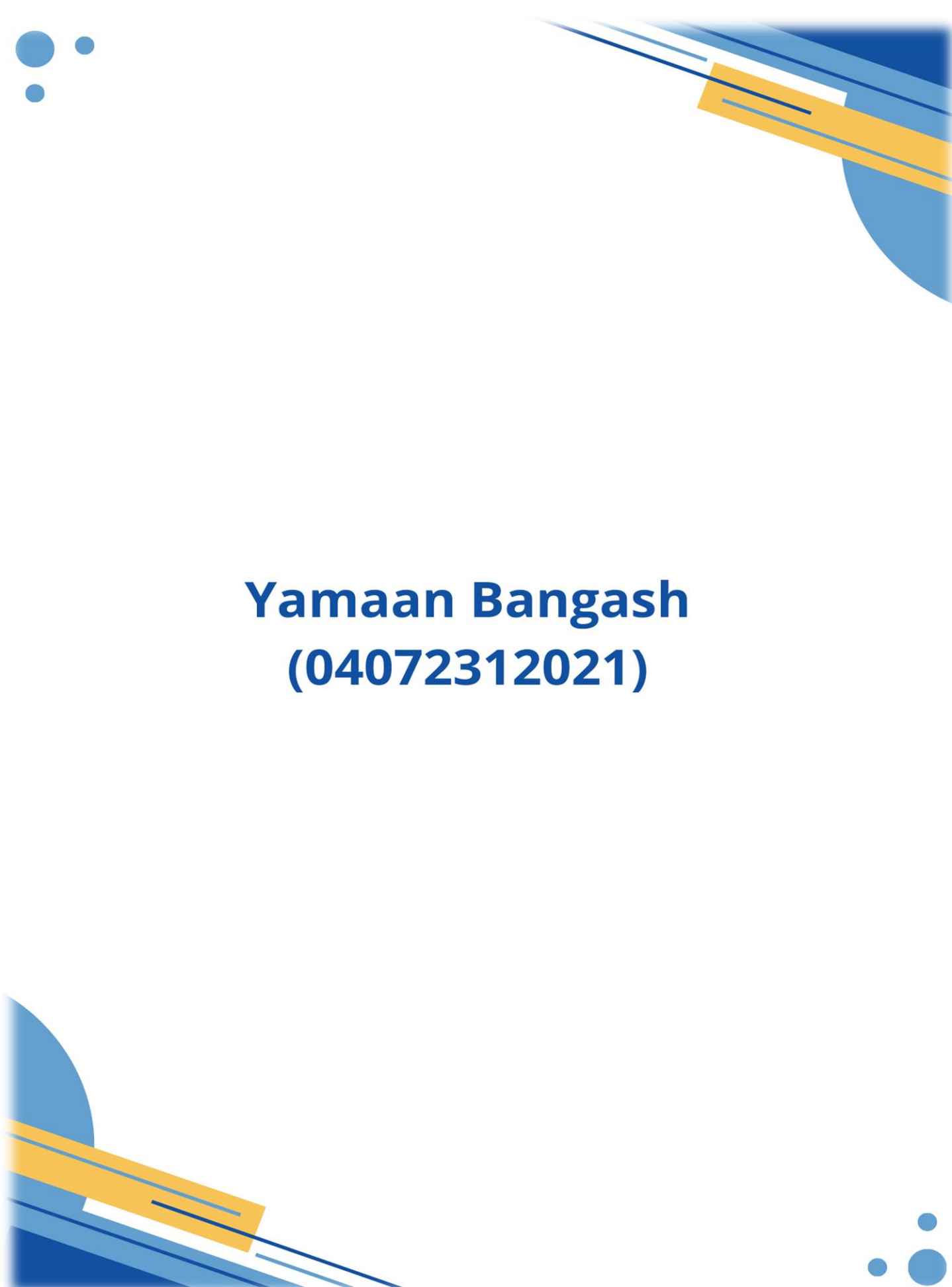
Cardinality: One-to-Many (1:M) → A coordinator can view multiple students.

9. Registration Record

Stores the details of student course enrollments.

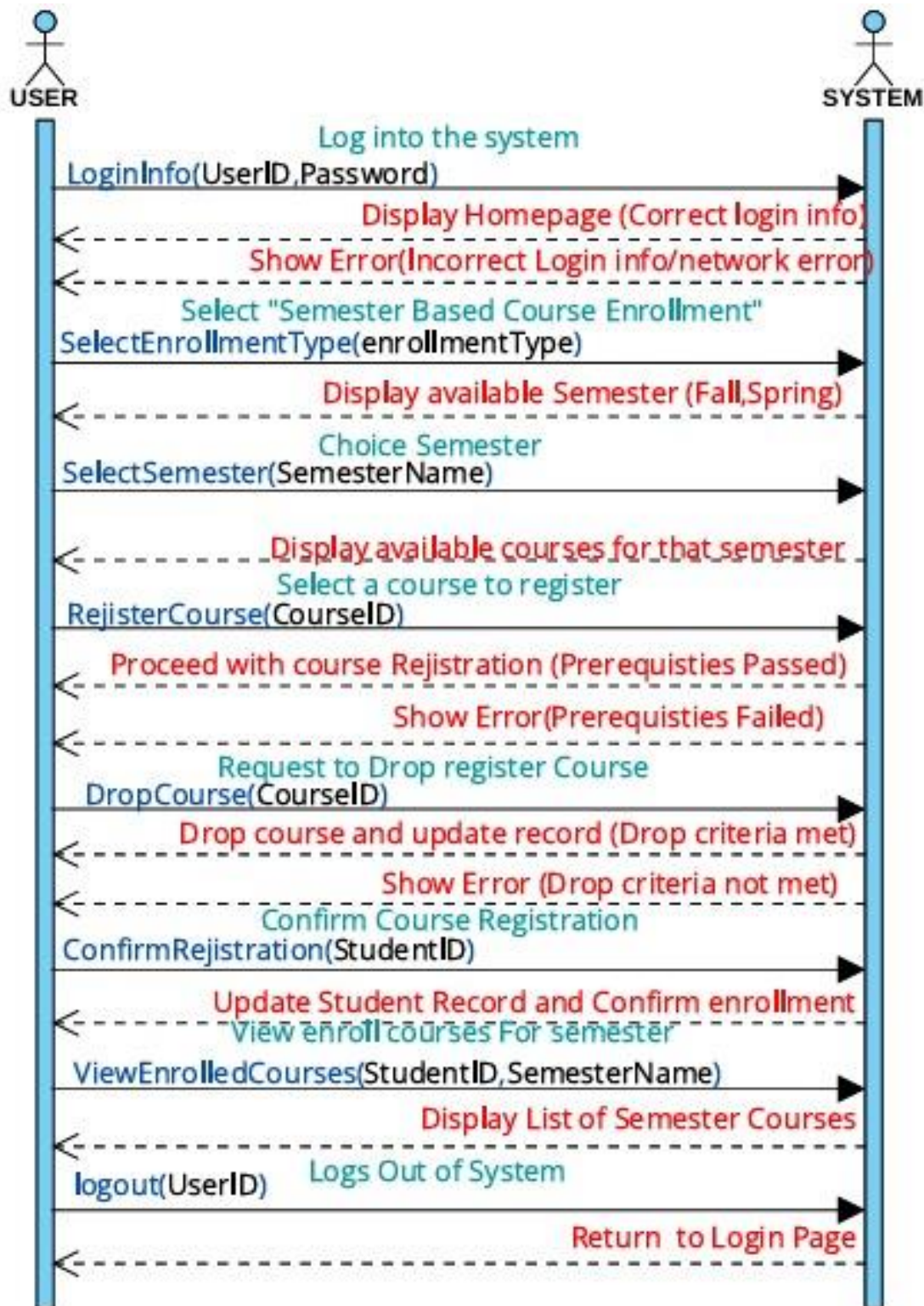
- **RegistrationID** (*VARCHAR, 10-15*) – Unique registration identifier.
- **StudentID** (*VARCHAR, 10-15*) – The student who registered.
- **CourseID** (*VARCHAR, 6-10*) – The course in which the student is enrolled.

Cardinality: One-to-Many (1:M) → A registration record can contain multiple courses.



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Semester Based Course Enrollment

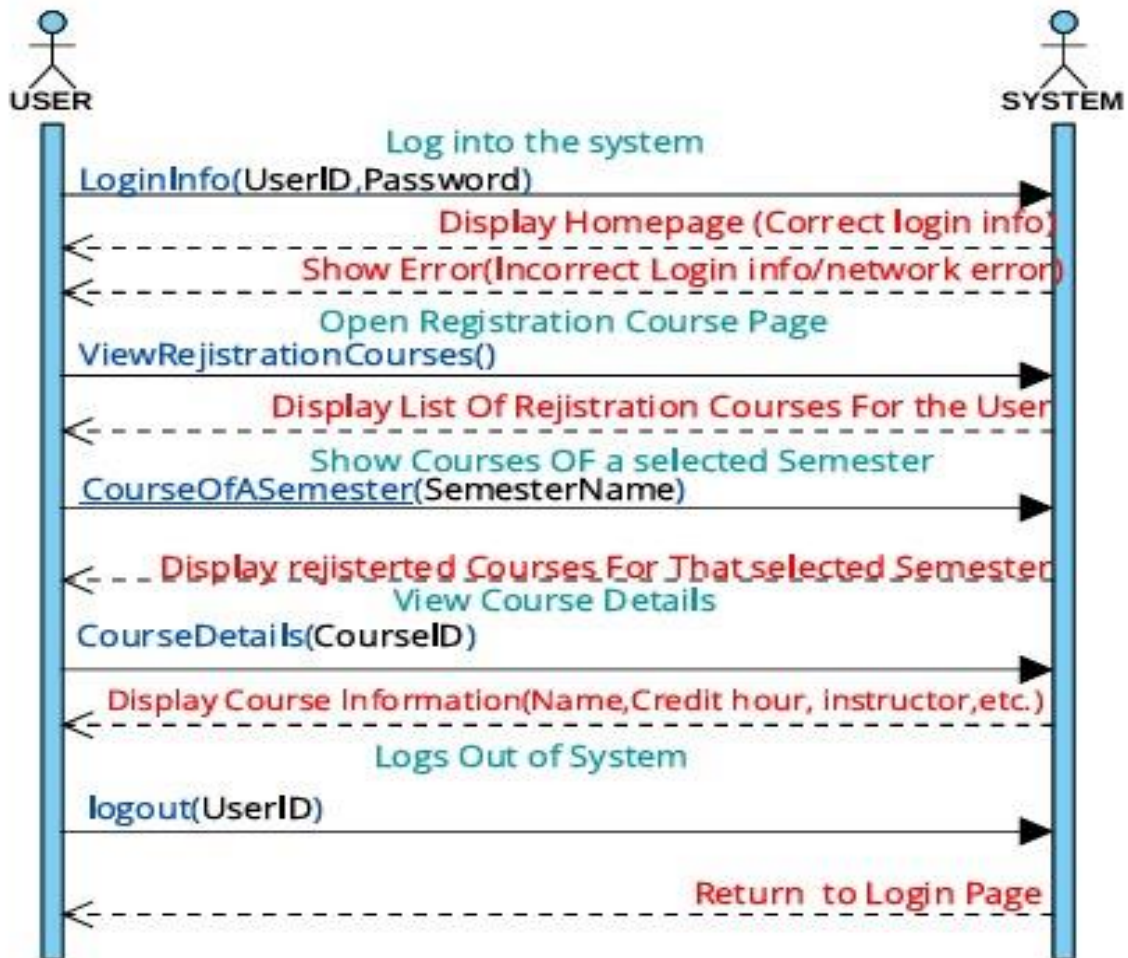


Manage Academic Record



View Registered Courses

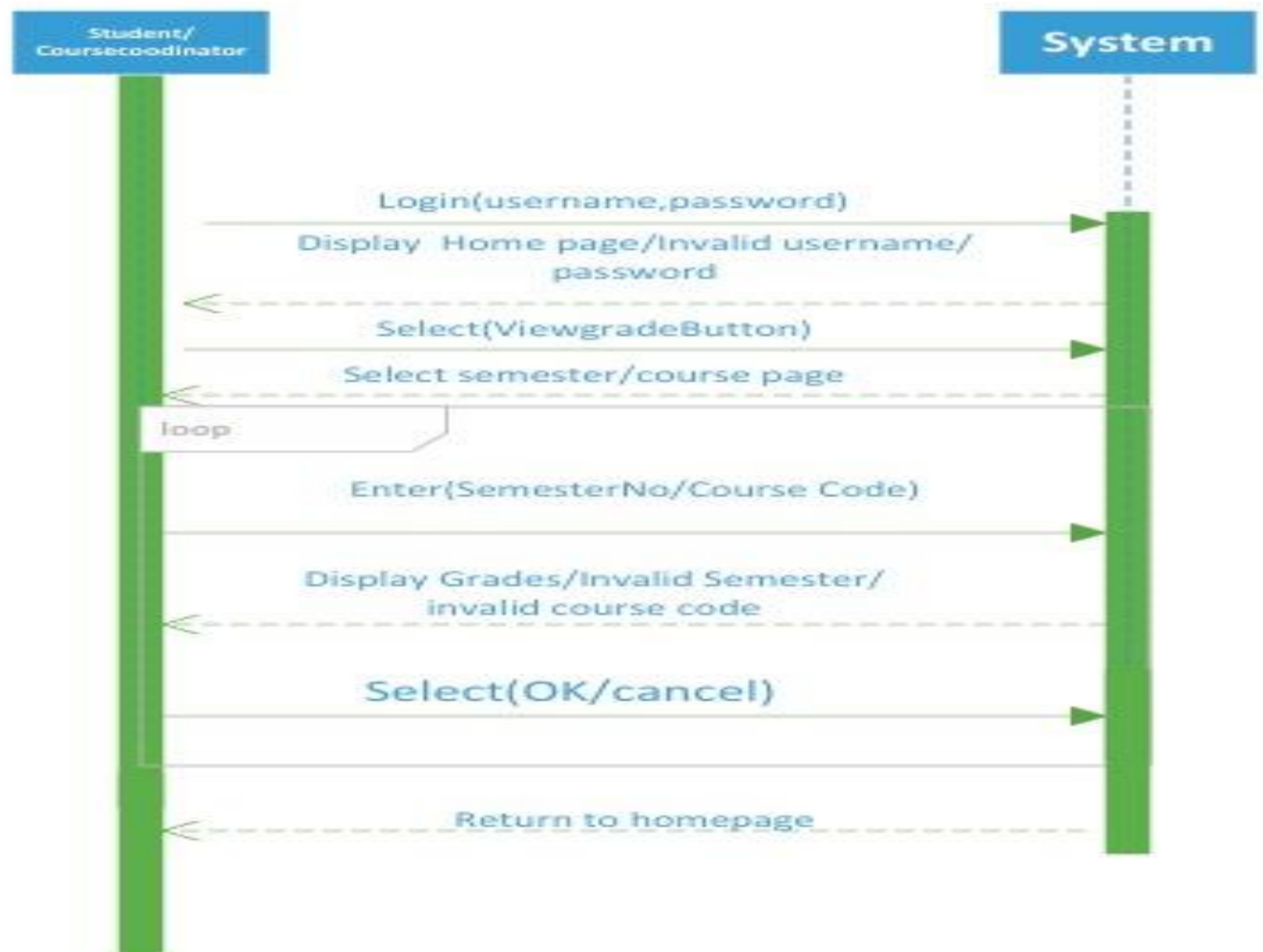
x



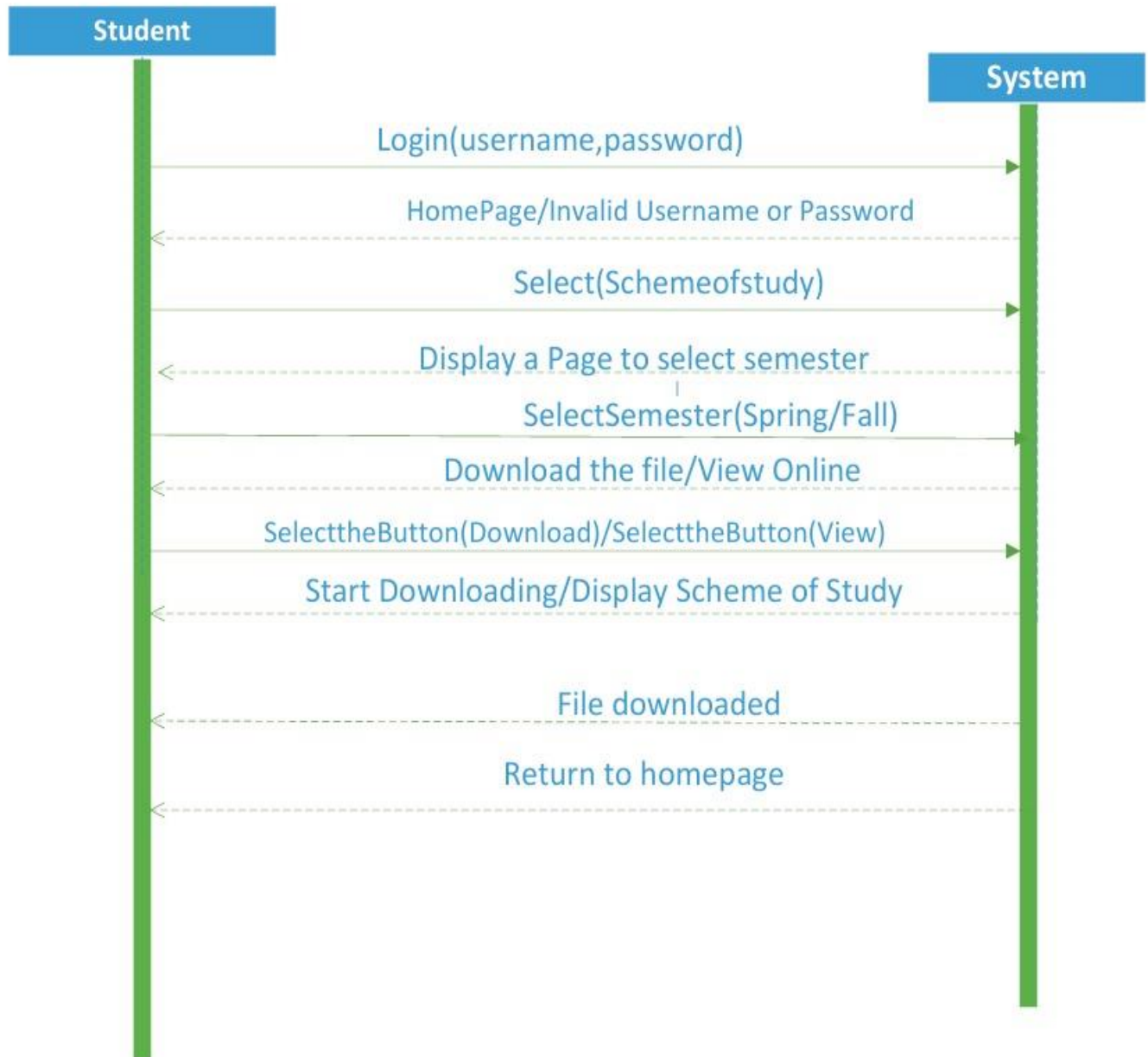


Ahsen Rasheed
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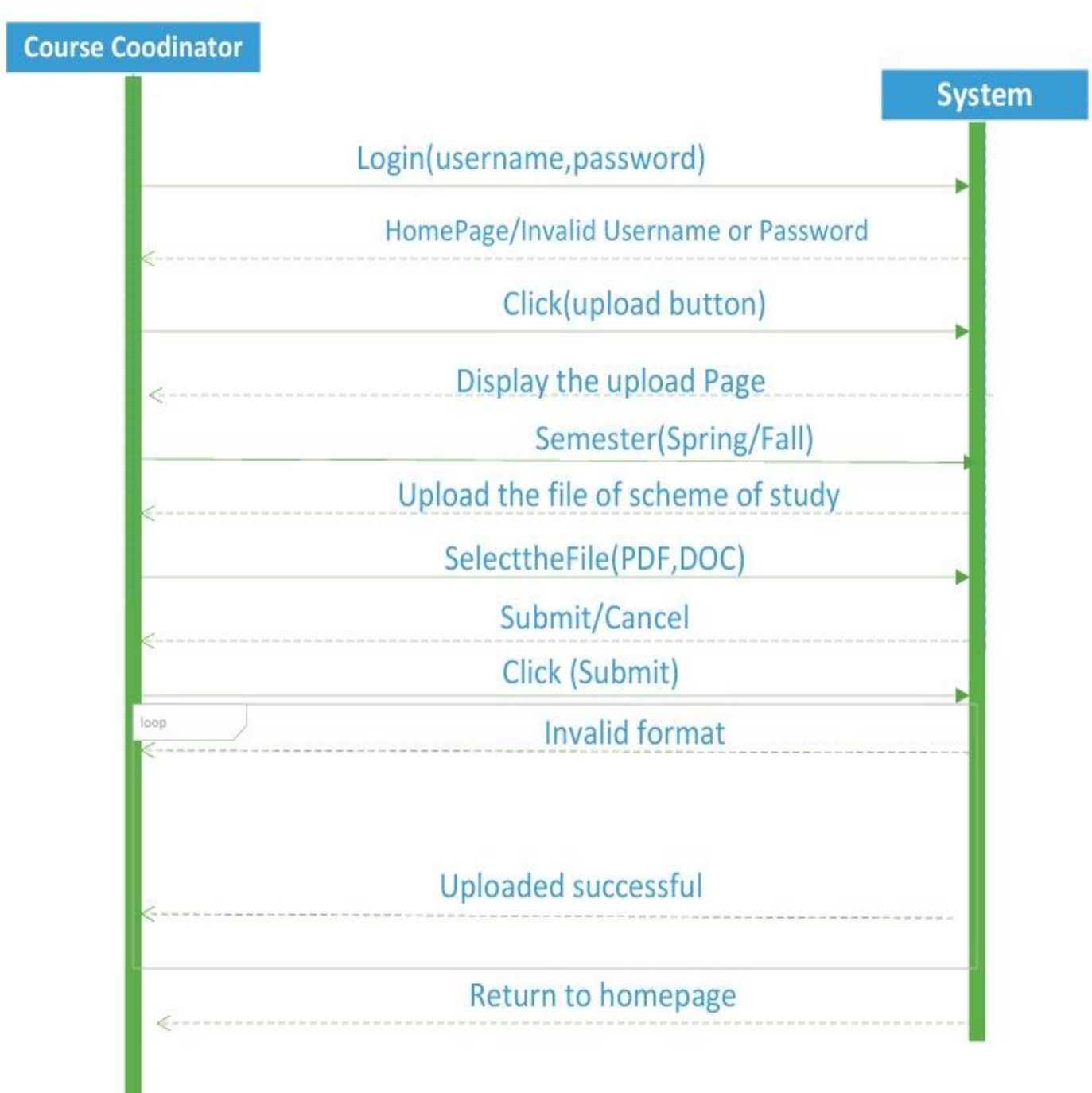
View Grade



View Study OF Scheme



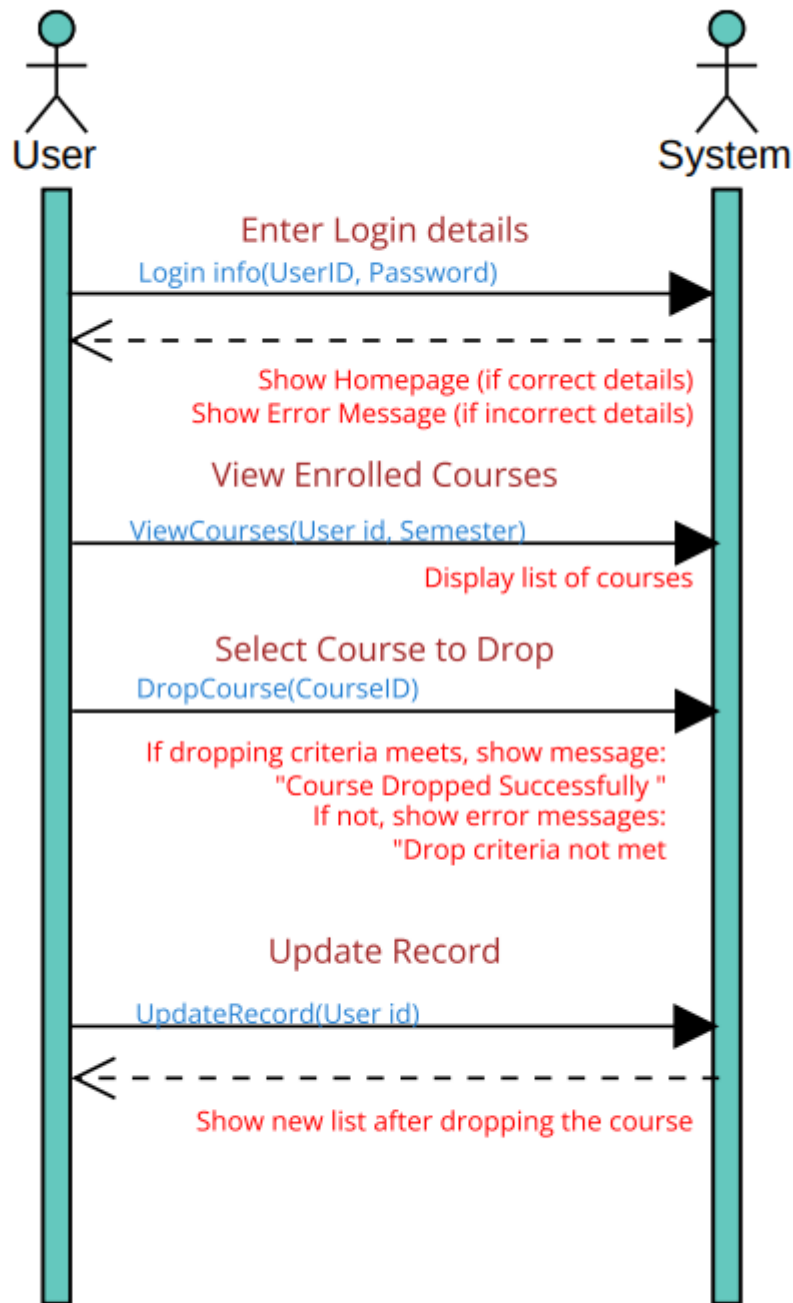
Upload Scheme OF Study





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Drop Course



Authentication

